

Due on Tuesday September 7 at 11:59 PM

In these exercises, all coupon bonds pay coupons twice per year and the coupon rate is given as an annualized rate..

- Exercises 1.2, 1.3, and 1.4 from Tuckman & Serrat written up together. (These exercises all concern the same list of bond prices.)
- Exercises 2.3 and 2.5 from Tuckman & Serrat written up together.
- The table below gives prices (per \$100 face) of three different bonds. Notice that two of the bonds pay coupons.

Maturity	Coupon Rate	Price
.5 years	.06	99.758
1 year	0	93.351
1.5 years	.08	100.734

- Find the discount factors $d(.5)$, $d(1)$, and $d(1.5)$.
 - Find the spot rate $\hat{r}(1)$ and the forward rate $f(1)$.
 - Find a coupon rate q^* such that the price of a bond with maturity 1.5 years, face value \$100 and coupon rate q^* will be equal to \$100.
- Coupon bonds with maturity $T = 10$ years, face value \$10,000, and annual coupon rate $q = .04$ are currently trading at \$9,060.00 per bond. Coupon bonds with maturity $T = 10$ years, face value \$5,000, and annual coupon rate $q = .08$ are currently trading at \$6,146.50 per bond. Determine the prices of each of the following fixed income securities.
 - A zero coupon bond with face value \$20,000 and maturity 10 years.
 - An annuity that will make payments of \$500 twice per year for the next 10 years.
 - A coupon bond with maturity 10 years, face value \$7,500 and annual coupon rate $q = .06$.
 - Assume that the spot rates for maturities .5, 1, 1.5, and 2 are given by $\hat{r}(.5) = .04873$, $\hat{r}(1) = .04496$, $\hat{r}(1.5) = .04782$, $\hat{r}(2) = .04891$. A customer walks into a bank today and agrees to borrow \$100,000 6 months from today and to repay the loan with a single lump-sum payment 2 years from today. How much will the customer pay the bank at time 2?

6. For $\eta, T > 0$ such that $T - \eta$ is a multiple of 3 months (i.e., a multiple of $\frac{1}{4}$ years), let $\tilde{r}_{0,\eta,T}^{for}$ be the forward interest rate, with quarterly compounding, agreed upon at time 0 for a loan to be initiated at time η and repaid with a single payment at time T . (Nothing is paid to enter into a forward loan agreement.) For every dollar borrowed at time η , the amount to be repaid at time T is

$$\left(1 + \frac{\tilde{r}_{0,\eta,T}^{for}}{4}\right)^{4(T-\eta)}.$$

Assume that $d(3) = .88888$ and $d(3.75) = .86322$. Determine $\tilde{r}_{0,3,3.75}^{for}$.

7. Let $T > \eta > 0$ be given and let $\hat{R}(\eta)$, $\hat{R}(T)$ denote the effective spot rates for dates η and T , and let $R_{0,\eta,T}^{for}$ denote the effective forward agreed upon at time 0 for borrowing between time η and time T . Recall that for every dollar borrowed at time η the amount to be repaid at time T is $(1 + R_{0,\eta,T}^{for})^{T-\eta}$. Assume that $\hat{R}(T) > \hat{R}(\eta)$. Show that

$$R_{0,\eta,T}^{for} > \hat{R}(T).$$

8. (Continuously Compounded Spot Rates) Let $\hat{r}_c(t)$ denote the spot rate for time t using continuous compounding so that

$$d(t) = e^{-t\hat{r}_c(t)}.$$

We assume that the function $\hat{r}_c : (0, \infty) \rightarrow \mathbb{R}$ is continuously differentiable. Determine a condition on the function $\hat{r}_c$ that is necessary and sufficient for the discount factors to satisfy

$$d(t_1) \geq d(t_2) \text{ for all } t_1, t_2 \in (0, \infty) \text{ with } t_1 \leq t_2.$$

(Your answer should involve $\hat{r}_c$ and the derivative $\hat{r}_c'$.)